

Haochen Hu

EDUCATION

- 2023-now** **Ph.D.**, *Department of AAE.*
The Hong Kong Polytechnic University (PolyU), Hong Kong, China.
Supervisor: [Prof. Chih-Yung Wen](#), **Co-Supervisor:** [Prof. Bing Wang](#)
- 2021-2023** **M.Sc., Mechanical Engineering**, *Department of ME.*
The Hong Kong Polytechnic University (PolyU), Hong Kong, China.
GPA: 3.7/4.3 (Outstanding Honor)
Supervisor: [Prof. Boyang Li](#)
- 2015-2019** **B.E, Aircraft Design and Engineering (Aerospace)**, *College of Astronautics.*
Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, China.
GPA: 3.5(85/100) **Ranks:** 5/33 (15.15%)

RESEARCH

- 2023-now *Dept. of AAE, PolyU.*
- Thesis: Promoting Visual Intelligence under Challenging Underwater Environment: From 2D to 3D Applications
- 2021-2023 *Dept. of ME, PolyU.*
- Thesis: Multispectral and Multi-type Feature Matching as Informative Guidance for Photogrammetry and Computer Vision
- 2019-2021 *Dept. of Automation, Tsinghua University.*
- Project: Perception & Kinetic simulation of self-driving motorbike
- 2017-2019 *College of Astronautics, NUAA.*
- Thesis: An active landform perception method based on visual saliency and SLAM for Planetary Rover exploration mission.

PUBLICATION

- **Hu, Haochen**, Bin, Yanrui, Wen, Chih-yung, & Wang, Bing. "RQUL-UIE: Revitalizing Quality-Unstable Labels for Underwater Image Enhancement via In-Dataset Self-Supervision." *Prepared for submission to IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2027).*
- **Hu, Haochen**, Bin, Yanrui, Zhang, Zhengyan, Wei, Minchen, Wen, Chih-yung, & Wang, Bing. "Fusing Transferred Priors and Physics-based Decomposition for Underwater Image Enhancement." *Information Fusion*, major revision (IF: 15.5, JCR Q1, CAS B1 TOP).
- **Hu, Haochen**, Lu, Yanze, Xue, Yuanbo, Zhang, Zhengyan, Yang, Shu, Wen, Chih-yung, & Wang, Bing. "NVS-UIE: Diffusion UIE towards Novel View Synthesis via Physics-Aligned Frequency Fusion." *Visual Intelligence*, minor revision.
- **Hu, Haochen**, et al. "Underwater sequential images enhancement via diffusion and physics priors fusion." *Information Fusion* 115 (2025): 103365. (IF: 15.5, JCR Q1, CAS B1 TOP)
- **Hu, Haochen**, Li, Boyang, Yang, Wenyu, & Wen, Chih-yung. "A novel multispectral line segment matching method based on phase congruency and multiple local homographies."

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Remote Sensing (2022). (IF:4.1, JCR Q1, CAS B2)

- Zhang, Zhengyan, **Hu, Haochen**, Wang, Bing, Lin, J., Li, J., Liu, M., & Wen, Chih-yung. "Pseudo Forward Depth Estimation for Imaging Sonar Using Diffusion Models." *International Conference on Information and Automation (ICIA)* (2025).
- Yang, Wenyu, Luo, Haojun, Tse, Kwai-wa, **Hu, Haochen**, Liu, Kang, Li, Boyang, & Wen, Chih-yung. "Autonomous-targetless extrinsic calibration of thermal, RGB, and LiDAR sensors." *IEEE Transactions on Instrumentation and Measurement* (2024).
- Yang, Wenyu, **Hu, Haochen**, Tse, Kwai-wa, Chen, S., Wen, W., & Wen, Chih-yung. "LiDAR stereo visual inertial pose estimation based on feedforward and feedbacks." *International Conference on Unmanned Aircraft Systems (ICUAS)* (2024).
- Leng, S., **Hu, Haochen**, Yu, M., & Ju, H. "Active manipulator motion planning for planetary landform awareness." *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering* (2019).

SELECTED GRANTS AND AWARDS

Time	Name	Granting Institution
2023	3+1 Research Student Scholarship	PolyU
2023	Outstanding Graduates Scholarship	Faculty of Engineering, PolyU
2019	Excellent Bachelor's Degree Thesis	<i>College of Astronautics, NUA</i>
2017	The Third Prize Scholarship	<i>NUAA</i>
2017	Second-class academic scholarship	<i>NUAA</i>
2015	Excellent Student Cadre	<i>NUAA</i>